USB-C Bench PSU

Rev 1.0

Compact Dual-Rail Power Supply | 3.3V & 5V | USB-C Input | Fully Analogue

The USB-C Bench PSU is a compact, open-source power supply board that converts USB-C 5V input into regulated 3.3V and passthrough 5V output rails. It requires no firmware or configuration; connect USB-C power and select your output rail with the onboard slide switch. Designed for breadboard prototyping and benchtop use.

Key Features

Fully analogue — No microcontroller, no firmware, instant-on operation

USB-C input — 5.1k CC pulldowns for proper 5V enumeration at up to 3A

Dual output rails — Regulated 3.3V (TPS563201 buck) + 5V passthrough

Output switching — P-channel MOSFETs controlled by SPDT slide switch

Input protection — Resettable PTC fuse (2A hold / 3.5A trip)

ESD protection — SMF5.0/6.0A TVS diodes on output rails

LED indicators — White LED (5V active), Red LED (3.3V active)

JLCPCB ready — All SMD parts on LCSC, top-side assembly only

Absolute Maximum Ratings

Parameter	Min	Max	Unit
Input voltage (VBUS)	-0.3	6.0	V
Input current (continuous)	—	3.0	A
Output current, 3.3V rail (continuous)	—	2.0	A
Output current, 3.3V rail (peak)	—	3.0	A
Output current, 5V rail (continuous)	—	2.0	A
Output current, 5V rail (peak)	—	3.0	A
Operating temperature	-20	+70	°C
Storage temperature	-40	+85	°C
ESD tolerance (HBM)	—	2000	V

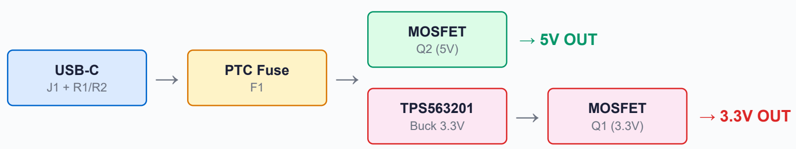
Electrical Characteristics

VIN = 5V, TA = 25°C, unless otherwise noted

Parameter	Min	Typ	Max	Unit
Input voltage	4.5	5.0	5.5	V
3.3V output (no load)	3.234	3.3	3.366	V
3.3V output (1A load)	3.17	3.24	3.3	V
3.3V output (2A load)	3.04	3.17	3.24	V
5V output (no load)	—	VBUS	—	V
5V output (2A load)	4.80	4.89	—	V
MOSFET Rds(on) Q1,Q2	—	53	85	mΩ
Switching frequency	—	560	—	kHz
PTC hold / trip current	—	2.0/3.5	—	A

Output voltages under load include voltage drop across Q1/Q2 MOSFET Rds(on). Feedback divider senses pre-MOSFET voltage; output drop is unregulated and load-dependent.

Functional Block Diagram



S1 (SPDT) controls Q1/Q2 gate drive

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Bill of Materials

Ref	Description	Value	Package	LCSC #	Qty
IC1	Synchronous buck converter	TPS563201	SOT-23-6	C116592	1
J1	USB-C receptacle (6-pin)	217175-0001	SMD	C668623	1
L1	Power inductor	3.3μH	1008	C5832373	1
Q1, Q2	P-channel MOSFET	AO3401A	SOT-23	C15127	2
F1	Resettable PTC fuse	2A / 3.5A	1206	C2685749	1
S1	SPDT slide switch	SS12D07VG4	TH	C2939726	1
D1	TVS diode	SMF6.0A	SOD-123FL	C2857264	1
D2	TVS diode	SMF5.0A	SOD-123FL	C2857263	1
D3	White LED	—	0603	C2290	1
D4	Red LED	—	0603	C2286	1
C4	Input capacitor	47μF	0805	C19103846	1
C2, C5	Decoupling capacitor	22μF	0603	C20416425	3
C1, C3, C6	Bypass/Decoupling capacitor	100nF	0402	C1525	3
R1, R2	CC pulldown resistor	5.1kΩ	0402	C25905	2
R3, R6, R7	Gate pull-up resistor	100kΩ	0402	C25741	3
R4	Feedback divider (top)	33.2kΩ	0402	C2930001	1
R5	Feedback divider (bottom)	10kΩ	0402	C25744	1
R8	LED resistor (5V rail)	330Ω	0402	C25104	1
R9	LED resistor (3.3V rail)	220Ω	0402	C25091	1
JMP1	Configuration jumper	—	2-pin TH	—	1

All SMD parts are available as JLCPCB basic or extended parts. S1 and JMP1 are through-hole.

⚠ Important Safety Warning

Disconnect loads before switching output voltage

Both MOSFET drains share a common output node. When the 5V rail is selected, Q1's body diode forward-biases and holds the 3.3V regulator output at approximately 4.3V (5V minus the body diode drop). When switching from 5V to 3.3V mode, the output will momentarily remain near 5V until the output capacitors discharge through the connected load. This transient can exceed the absolute maximum voltage rating of 3.3V devices and cause permanent damage.

Always disconnect or power-off loads before changing the output voltage selection. Switching from 3.3V to 5V mode does not have this issue, as Q2's body diode is reverse-biased in that direction.

Application Information

USB-C Compatibility

The board uses 5.1kΩ CC pulldown resistors (R1, R2) for USB-C enumeration. This requests the default 5V power profile without PD negotiation. Works with any USB-C charger or host port. For reliable operation, use a USB-C to USB-C cable. **NOTE:** some cheap USB-C to USB-A cables omit CC wires, which may limit available current to 500mA.

3.3V Regulation

The TPS563201 operates as a synchronous buck converter at ~560kHz with internal compensation. The feedback divider (R4 = 33.2kΩ, R5 = 10kΩ) sets the output: $V_{out} = 0.765V \times (1 + R4/R5) = 3.30V$. The 3.3μH inductor (L1) and output capacitors provide filtering and energy storage.

Output Switching

Both output rails are gated by AO3401A P-channel MOSFETs (Q1, Q2). The SPDT slide switch (S1) drives the MOSFET gates via 100kΩ pull-up resistors (R3, R8, R9). When a rail is disabled, the gate is pulled high ($V_{gs} = 0V$) and the MOSFET is off.

Protection

F1 is a resettable PTC fuse on the VBUS rail protecting against short circuits and overcurrent faults. D1 and D2 are unidirectional TVS diodes clamping transients on the output rails. The TVS clamp voltage is 6.4V (typ) at 1A.

LED Indicators

D3 (white, R6 = 330Ω) indicates the 5V rail is active. D4 (red, R7 = 220Ω) indicates the 3.3V rail is active. Both LEDs draw approximately 5mA at their nominal operating points.

Footprint

Board footprint is extremely compact, fitting in a 22mm x 17mm rectangle.

Ordering & License

 Design files, Gerbers, BOM, and pick-and-place data are available at:

github.com/TheRealEuaX/usb-c-bench-psu

Manufactured boards for purchase coming soon.